

UTM TENSILE REPORT — UTM_Test_20260814_120352_100%infill_Videocapture

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

46.5 MPa

UTS

40.5 MPa

Yield σ_y

2.93 GPa

Modulus E

17.5 %

Failure ϵ_f

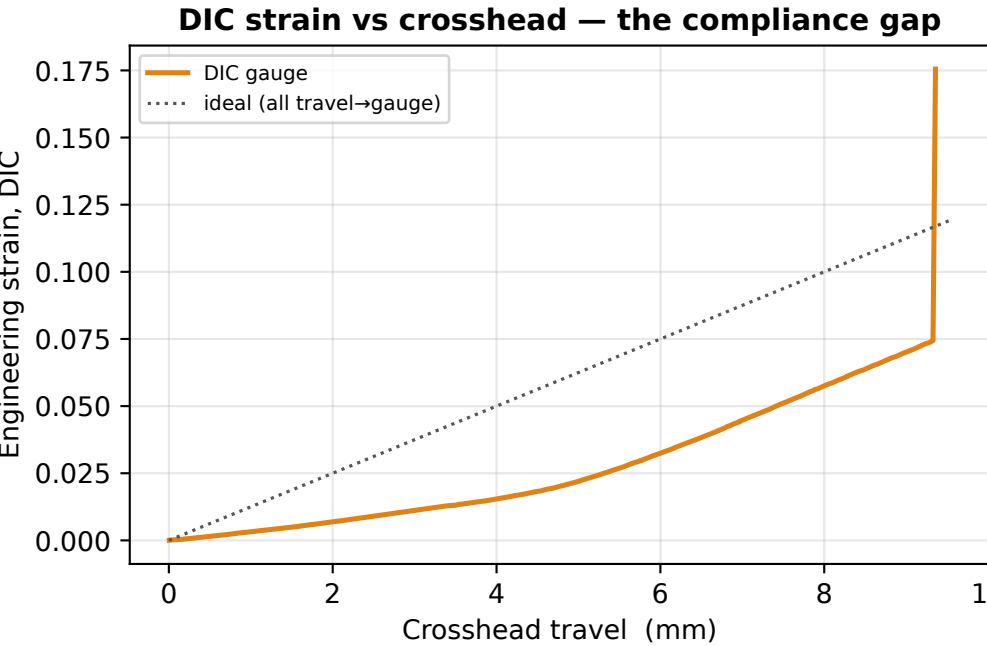
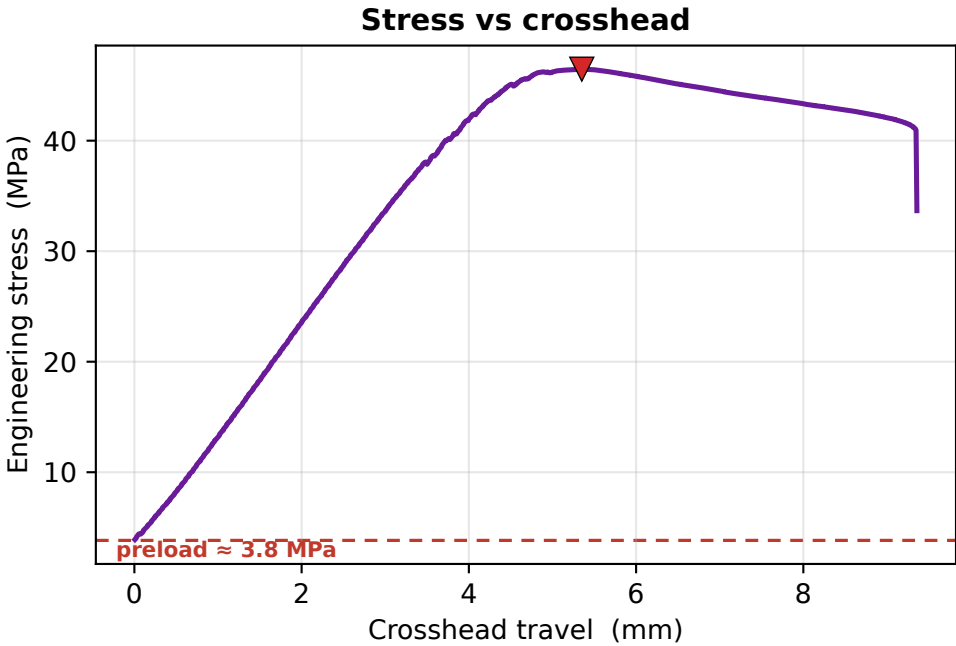
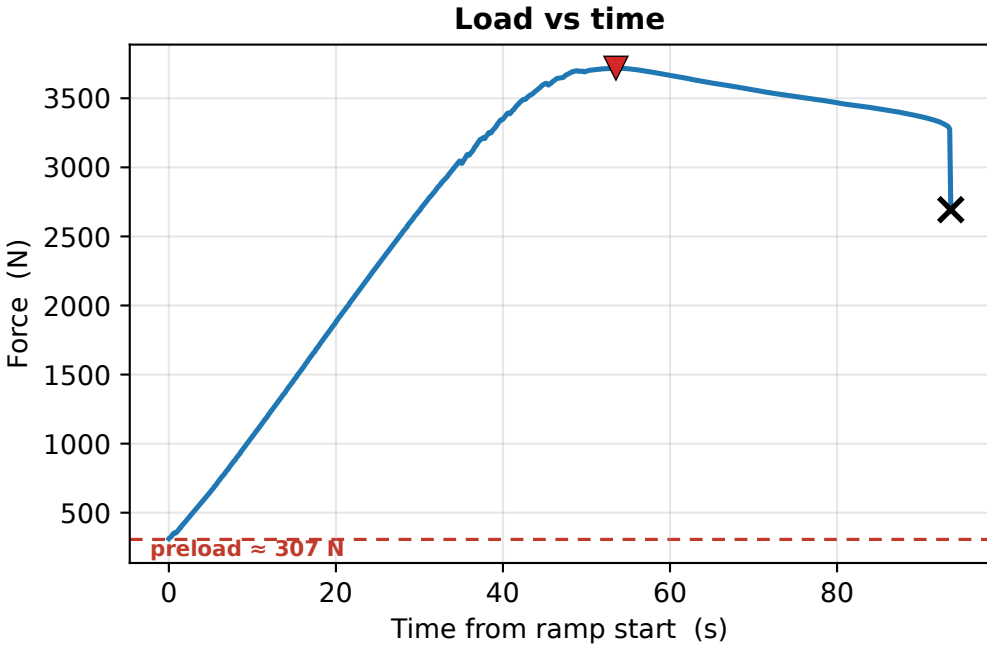
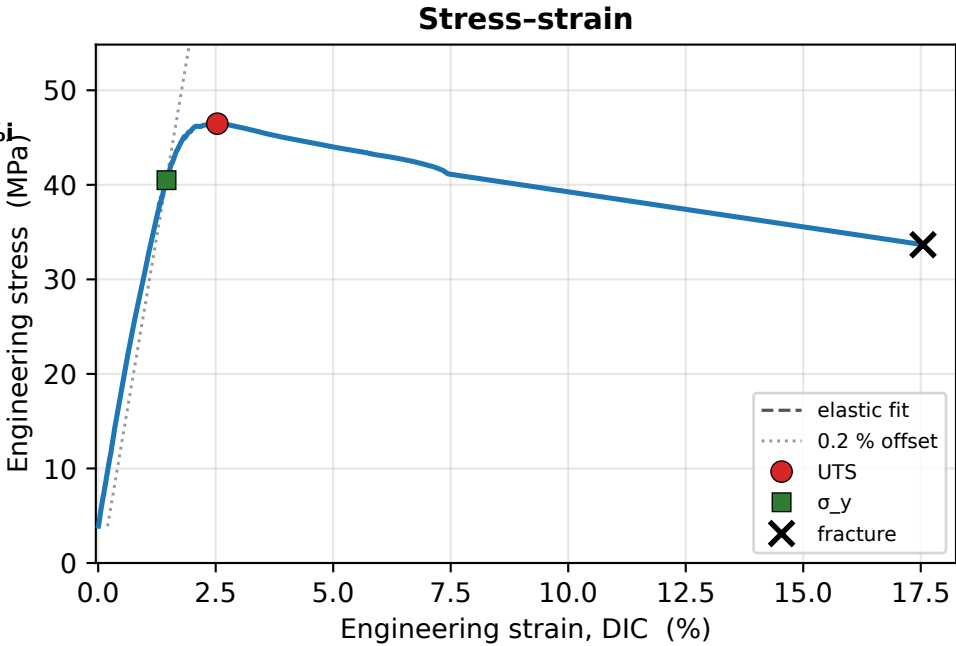
6774 kJ/m³

Toughness

307 N

Preload anchor

TEST SETTINGS	
Specimen	UTM_Test_20260814_120352_100%infill
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 331.461
Comment	—
Test date	2026-08-14 12:01:18
Duration	188.2 s



VALIDATION vs REFERENCES			
prop	meas	Chacón	E-PLA k
UTS	46.5	32-60 PASS	×1.25
σ_y	40.5	30-50 PASS	×1.43
E	2.9	3-5.5 low	×0.98
ϵ_f	17.5 %	(range 3-7 %)	×0.46
✓ strength inside journal range (k≈1)			